

Madhusudan Abhay Ambulkar

+49 15510820982 | madhusudanambulkar@gmail.com | Frankfurt am Main, Hesse, Germany | [LinkedIn](#) | [GitHub](#)

SKILLS

Data & Analytics: Power BI (DAX, Time Intelligence, KPI Dashboards), SQL, Data Mining, Data Cleaning & Transformation, Statistics, ETL Pipelines

Programming & Tools: Python (Pandas, NumPy, scikit-learn), Excel (VBA/Macros), Jupyter Notebook, Git/GitHub, VS Code, Google Colab.

Gen AI & AI Agents: YOLO (object detection, edge deployment), LLM APIs (Claude, OpenAI), RAG, LangChain, Prompt Engineering, Power Automate.

Dashboards & Reporting: Streamlit, MS Office Suite (Excel, PowerPoint, Word), SOP Writing, Atlassian JIRA.

Languages: English (C1 - IELTS 7.0), German (A1.2, actively improving)

WORK EXPERIENCE

eClinicalWorks

Mumbai, India

Software Specialist

July 2022 – August 2024

- Queried and explored **relational databases**, connecting to **MySQL** and to **run ad-hoc SQL queries, inspect schemas**, and validate **data integrity** across healthcare billing and operational records.
- Used **SQL queries and database tools** to investigate and resolve **300+ data and system issues**, improving operational resolution **efficiency by 20%** in an agile team environment.
- Documented and optimized** internal **sales and operational processes (SOPs)**, reducing incident **escalation time by 15%**; authored clear process documentation accessible to non-technical stakeholders across teams.
- Collaborated with **Sales and Product Analysis** teams to design **data visualizations** in **Power BI, SQL, and Excel**, translating complex data analysis into clear, actionable reports and visualizations for non-technical stakeholders across multiple teams.

PROJECTS

Distributed Raspberry Pi Cluster & AI Intruder Detection System

- Built a **multi-node hardware cluster** using a **Raspberry Pi** master and 8 worker nodes via an **automated PXE network** boot server for diskless OS deployment.
- Trained a **YOLOv8n object detection model** on 34,000+ images, achieving a 0.758 mAP for security threats, and ran it on a **Raspberry Pi edge node**.
- Set up an **edge-to-cloud data pipeline** using an **MQTT broker, custom ETL logic**, and **Kubernetes** to manage real-time inference workloads.

Real Time Air Quality Monitoring System - Python, Zephyr RTOS

- Designed and maintained a structured **6-stage ETL pipeline** processing **multi-sensor environmental data** with **99.8% packet integrity** and a mean end-to-end **latency of 29.7ms**.
- Implemented **compliance classification logic** aligned with **ASHRAE 62.1 and WHO standards**, automatically evaluating **fused sensor data** against regulatory thresholds and classifying environmental severity into four tiers (OK → CRITICAL) for real-time health advisories.
- Built a **Python Streamlit analytics dashboard** delivering sub-second visual updates for non-technical users; also authored full **system documentation** covering architecture, data flows, and sensor specifications.

EDUCATION

MS in High Integrity Systems (AI & Data Science)

Frankfurt University of Applied Sciences

Frankfurt am Main, Germany

October 2024 – present

Bachelor of Science in Information Technology

University of Mumbai

Mumbai, India

2019 – 2022